Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of a modular intelligent spatial safety platform based on PINNs and on-device AI**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials			
	Medium Category	Small Category	Medium Category	Small Category
	Industrial/ General Machinery	*Fire Safety Machinery*		
Concept of Proposed Technology	**Core Concept** An on-device AI–based modular spatial safety platform that integrates a physics-informed (PINN-based) real-time fire detection and prediction engine with an intelligent Space Operating System (Space OS), maximizing both operational efficiency under normal conditions and survival rates in emergency situations.			

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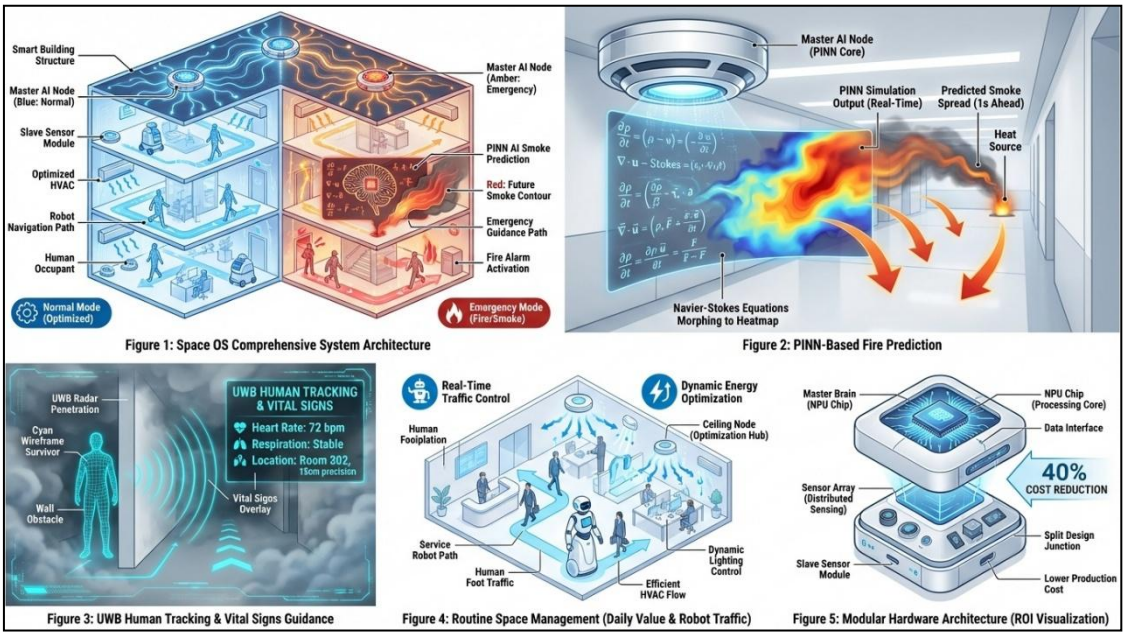

Figure. Concept overview of modular intelligent spatial safety platform

Key Functions and Potential Applications

Detection and Prediction Stage

1) Development of a PINN (Physics-Informed Neural Networks)–based surrogate model
   - Establish a physics-informed fire prediction capability that enables robust estimation of smoke and heat propagation, even in complex and previously unseen building environments.
   - Support near-real-time inference suitable for time-critical decision-making without reliance on centralized computation.
2) Physics-based triggering and false-alarm filtering
   - Develop early fire detection mechanisms that combine physical cues and intelligent analysis to improve detection reliability.
   - Enable robust suppression of false alarms through cross-validation of fire-related patterns, reducing malfunctions caused by non-fire events.
3) Human location and biometric information fusion support technology
   - Enable high-precision indoor localization of occupants in fire-affected environments.
   - Support identification of high-risk occupants through non-contact assessment of safety-related physiological indicators.
   - Ensure privacy-preserving data processing through localized analysis of sensitive information.
4) Robot–Space integrated infrastructure for disaster response
   - Establish a robot–space integrated infrastructure that enables coordinated operation of heterogeneous robots and spatial elements during emergencies.
   - Support real-time awareness of occupant and robot states to enhance

situational understanding.

- Ensure reliable communication and control of critical alerts and response commands under adverse disaster conditions.

Operation and Control Stage: On-Device Spatial OS

1) Normal Mode (Everyday Spatial OS)
   - Enable real-time monitoring of occupant and robot movement to support unified spatial management.
   - Support intelligent energy and space utilization based on occupant distribution and movement patterns.
   - Provide indoor traffic and congestion management to improve safety and mobility efficiency.
   - Enable continuous monitoring of safety-related anomalies, including non-fire incidents.
2) Emergency Mode (Emergency Spatial OS)
   - Enable real-time situational awareness through fusion of fire prediction and sensor information.
   - Support dynamic evacuation route guidance based on occupant location and mobility.
   - Enable crowd flow optimization and adaptive building control during emergency situations.
   - Support robot-assisted early-stage fire response to mitigate fire growth during the initial response phase.
3) Emergency Survival Guidance
   - Enable non-contact identification of trapped occupants and assessment of survival status under low-visibility conditions.
   - Support real-time sharing of occupant location and condition with emergency response centers.
   - Provide optimized rescue access route guidance to support efficient rescue operations.

Service Scalability

1) Non-contact, Privacy-Preserving AI-Based Gait Analysis for Fall Detection
   - Enable non-contact and privacy-preserving monitoring of occupant movement patterns to detect fall risks.
   - Support early warning mechanisms for caregivers and facility staff upon detection of elevated risk.
2) Service Robot Integration and Initial Response
   - Enable robot-assisted patrol and rapid response to safety-related incidents.
   - Support on-site monitoring and assistance by service robots until human

Rationale for Joint R&D Project	responders arrive. Core Targets 1) Multi-story buildings, large-scale mixed-use shopping malls, senior care and nursing facilities, and smart city infrastructure
	Critical Technical Barriers 1) Engineering Critical Points ● Limitations of threshold-based detection: Conventional fire detection systems rely on sensor-specific thresholds, resulting in frequent false alarms and an inability to identify the initial characteristics of real fires (e.g., heat and flow dynamics), often causing the loss of the critical golden time. ● Lack of real-time capability in FDS simulations: While Fire Dynamics Simulator (FDS) is the standard for fire modeling, it requires massive computational resources, making real-time prediction infeasible. As a result, it cannot predict the dynamic physical propagation paths of smoke and flames during fires, limiting responses to static evacuation guidance. ● Loss of physical networks: Centralized server-based architectures are vulnerable to high temperatures and flames, which can destroy building communication cables and power networks during fire spread, leading to system-wide failure and preventing independent intelligent decision-making under extreme conditions. ● Passive idle infrastructure: Existing fire detection systems remain idle for 99.9% of the time, consuming power without delivering operational value until a fire occurs, resulting in extremely low return on investment (ROI). 2) Safety Requirements ● Technologies capable of predicting fire spread paths within 1 second after fire detection, and non-contact detection of vital signs (respiration and heart rate) of individuals trapped in smoke-filled environments. **Why Global Collaboration?** 1) Engineering Integration of Physics-Informed AI (PINN) and Hardware Optimization Technologies ● Interaction between data and physical laws: Conventional fire simulations (FDS) offer high accuracy but lack real-time capability, while general deep learning models provide fast inference but insufficient physical plausibility. ● Core technology from research institute: The adoption of PINN algorithms that incorporate world-leading fire physics datasets and embed the Navier–Stokes equations into the loss function, developed by research institute—a global leader in fluid dynamics—is essential. 2) Securing Global Market Leadership through Early Alignment with North American Fire

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	Safety Standards (NFPA/UL/FM) ● Certification barriers: North American standards such as NFPA, UL, and FM, which serve as global benchmarks for fire safety, impose highly stringent technical requirements, creating significant limitations for independent market entry. ● Strategic value of joint research: Through technical standard advisory support from research institute and validation using U.S.-based testbeds, joint research enables the acquisition of core technologies that comply with global standards from the initial design stage. 3) Transforming Idle Infrastructure Value to Address Social Challenges and Expand Profitability ● Innovation in return on investment (ROI): To overcome the limitations of passive fire detection infrastructure that remains idle for 99.9% of the time, the platform is expanded into a multi-purpose system capable of intelligent energy management and real-time tracking of occupants and robots. ● Balancing public interest and business viability: The solution simultaneously achieves public safety goals—such as building safety networks for aging societies—while delivering an always-on safety platform that is attractive to insurers and building operators, thereby maximizing market adoption and scalability.
Key Words	1. Physics-Informed Neural Networks (PINN) 2. On-Device AI (NPU) 3. Space Operating System (Space OS) 4. UWB Bio-Radar

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	**R&D Objectives** 1) Development of a Real-Time Fire Spread Prediction Capability ● Establish a real-time fire spread prediction capability that enables physically consistent estimation of smoke, heat, and flame propagation, even in complex and previously unseen spatial environments. ● Achieve ultra-fast prediction performance suitable for time-critical emergency response, enabling near-instantaneous generation of fire spread information without reliance on centralized computing infrastructure. 2) Development of High-Accuracy Fire Detection with Robust False-Alarm Suppression ● Develop a fire detection capability that provides high detection accuracy based on visual fire characteristics, ensuring reliable identification of actual fire events. ● Minimize false alarms by enabling robust discrimination between fire and non-fire events, addressing long-standing reliability issues of conventional fire detection systems 3) Development of a Data-Integrated Intelligent Spatial Operation Framework ● Establish an intelligent spatial operation framework that supports adaptive operation modes, enabling efficiency-oriented spatial management under normal conditions and dynamic, safety-driven decision-making during emergency situations. ● Enable non-contact identification of occupant location and safety status, supporting timely linkage to emergency response services to improve life-saving outcomes during the critical response window. 4) Modular Hybrid Hardware Platform ● Develop a modular system architecture that supports scalable deployment and flexible configuration according to building size and operational requirements. ● Ensure disaster-resilient operation, maintaining autonomous functionality and inter-node communication even under infrastructure damage or network disruption caused by fire. 5) Real-Time Dynamic Evacuation Route Optimization and Guidance ● Develop real-time evacuation guidance capabilities that dynamically identify and avoid hazardous zones, providing occupants with safer evacuation routes under evolving fire conditions. ● Enable adaptive evacuation guidance and crowd dispersion, preventing congestion at specific exits and improving overall evacuation efficiency through informed route distribution.			

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R&D Target Specifications	● Fire Prediction: Achieve real-time fire spread prediction capability suitable for time-critical emergency response. ● Prediction Accuracy: Ensure a high level of prediction reliability through consistency with physics-based reference models. ● Fire Detection: Provide accurate and dependable fire detection under diverse environmental conditions. ● False-Alarm Suppression: Significantly reduce false alarms by reliably distinguishing fire events from non-fire situations. ● Positioning: Enable high-precision indoor localization of occupants, even in adverse environments such as smoke-filled spaces. ● Evacuation: Support rapid computation of dynamic evacuation routes under large-scale occupancy conditions. ● Rescue (Vital): Enable reliable non-contact detection of occupant safety status to support timely rescue operations. ● Communication: Ensure low-latency and robust information exchange for safety-critical messages during emergencies. ● Robot Integration: Enable timely and coordinated response of heterogeneous robotic systems in emergency situations. ● System Reliability: Ensure highly reliable delivery of critical emergency messages under degraded network conditions.
R&D Plan	**Research Approach** 1) Physics–Data Hybrid Approach ● This project adopts a physics–data hybrid research approach to overcome the limitations of purely data-driven AI under dynamic fire conditions, ensuring physically consistent, robust, and generalizable prediction performance across diverse and previously unseen spatial environments. 2) Edge-Centric Intelligence ● An edge-centric system design is pursued to enable local execution of safety-critical functions, ensuring autonomous and continuous operation of the system even under communication disruptions or infrastructure damage during disaster situations. **Key Tasks and Methodologies** 1) PINN-Based Ultra-Fast Fire Propagation Prediction Engine ● Develop a real-time fire propagation prediction capability that provides rapid and physically plausible estimation of fire dynamics, supporting time-critical emergency decision-making. ● Establish methodologies to translate high-fidelity reference data and physical knowledge into computationally efficient predictive models suitable for real-time operation.

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	2) Vision-Based Hybrid Fire Detection and False-Alarm Filtering ● Develop fire detection methodologies that leverage visual and environmental cues to accurately identify fire events. ● Establish robust false-alarm suppression strategies to reliably discriminate between fire and non-fire situations, improving system trustworthiness under real-world operating conditions. 3) UWB-Based Precise Localization and Biometric Signal Fusion ● Use of UWB Time-of-Flight (TOF) measurements to achieve location accuracy of approximately 20 cm, even in dense smoke environments. 4) Space OS and Integrated Control of Heterogeneous Robots ● Develop an integrated operational framework that enables coordinated interaction between robots, occupants, and spatial infrastructure in both normal and emergency conditions. ● Support adaptive task allocation and movement coordination of robotic agents to enhance mobility efficiency, collision avoidance, and emergency response effectiveness. 5) Data-Driven Dynamic Evacuation Support Methods ● Develop data-driven evacuation support methodologies that dynamically adapt evacuation guidance based on evolving hazard conditions and occupant distribution. ● Enable cooperative operation between guidance systems and robotic agents to improve evacuation safety and effectiveness, particularly under low-visibility conditions.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institution plays a key role in ensuring the engineering reliability of the system by providing fundamental algorithm design grounded in fire physics and high-fidelity simulation data. 1) Technical Contributions ● Development of physics-informed modeling approaches for reliable fire behavior prediction. ● Establishment of physically consistent prediction methods applicable to complex spatial environments. ● Contribution to robust AI-based fire detection methodologies. ● Support for reference dataset construction reflecting realistic evacuation and mobility conditions. 2) Research Scope ● Development of representative fire and spatial reference scenarios for model validation. ● Definition of model optimization strategies suitable for real-time operation. 3) Resources Provided ● Provision of advanced research infrastructure and benchmark datasets. ● Technical advisory support for alignment with global fire safety standards. 4) Detailed Tasks ● Validation of fire behavior models across diverse spatial conditions. ● Establishment of a theoretical foundation supporting dissemination and standardization.
Expected R&R of Korean SME Partner	Korean SME Partner leads commercialization by optimizing and implementing core technologies in on-device environments, developing the hardware platform, and conducting real-world validation. 1) Technical Contributions ● On-device NPU optimization and lightweight deployment of PINN algorithms and fire-related AI algorithms. ● Development of hybrid validation logic that fuses vision-based fire detection AI with physical sensor data. ● Multi-object tracking: Individual identification and real-time tracking of occupants and robots using UWB and vision-based AI. 2) Development Tasks ● Intelligent Space OS: Development of middleware for energy management under normal conditions and dynamic evacuation guidance in emergencies. ● Real-time path optimization engine: Implementation of execution logic that reroutes robots based on occupant density or converts robots into guidance agents during fires. ● Distributed hardware architecture: Development of modular high-performance master nodes and slave nodes. ● Advancement of trapped occupant localization and vital sign detection algorithms using UWB and non-contact radar technologies. 3) Testing and Validation ● Conducting field demonstration tests at real target facilities such as multi-story

	buildings and large-scale shopping complexes. ● Validation of mesh network recovery and evacuation guidance system performance under simulated disaster scenarios. 4) Resource Investment and Collaboration ● Establishment of mass production frameworks for domestic and international market entry, and development of interfaces linked to the 119 emergency control system. ● Completion of globally standard-compliant product lineups through regular technical exchanges with research institutions.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project aims to secure next-generation intelligent spatial safety technologies by integrating Purdue University's physics-based fire prediction AI core technologies with Korean SME Partner's on-device Space OS platform, while generating practical commercialization outcomes and broad industrial and societal impacts. **Technical Outcomes** 1) Ultra-fast on-device PINN prediction engine 2) Physics-based hybrid fire detection 3) ROS2-based spatial OS for heterogeneous robot integration 4) Privacy-preserving non-contact precise localization **Economic Outcomes** 1) Entry into the global fire safety market 2) Space OS–based licensing and subscription (SaaS) business 3) Building operation efficiency and cost reduction (ROI) 4) Expansion into vertical markets such as senior care **Societal Impact** 1) Transformative reduction of disaster damage 2) Paradigm shift in the fire safety industry 3) Enhanced disaster resilience 4) Digital inclusion and social safety nets

6. References

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